



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

1. Data Collection

- Retrieved SpaceX launch records using **API calls** and **web scraping**.
- Consolidated raw mission, booster, and landing data into a unified dataset.

2. Data Wrangling & Cleaning

- Standardized column names, handled missing values, and corrected inconsistent labels.
- Converted data types (dates, numeric fields) and engineered features such as landing class, booster version, and payload categories.

3. Exploratory Data Analysis (EDA)

- Investigated relationships between payload, launch site, booster version, and landing outcomes.
- Identified key patterns: heavier payloads reduce landing success; certain sites have higher success ratios; booster reuse improves outcomes.

4. Interactive Visualizations

- Built a **Plotly + Dash dashboard** with:
 - Success-count pie chart for all sites
 - Success-ratio pie chart for the best site
 - Payload vs. outcome scatter plot with dynamic range slider
- Created a **Folium map** marking all launch sites with geographic context.

5. Predictive Modeling

- Built a classification model to predict booster landing success.
- Tested multiple algorithms (Logistic Regression, Decision Tree, Random Forest, SVM).
- Tuned hyperparameters and selected the best model based on accuracy.

Introduction

Project Background & Context

SpaceX Booster Landing Prediction

SpaceX relies on reusable rocket boosters to dramatically reduce launch costs. A successful landing allows a booster to be reused, saving **millions of dollars per launch** and improving turnaround time.

Each failed landing results in the loss of a high-value asset, increasing mission cost and operational risk.

Historical launch data contains patterns in payload, launch site, booster version, and flight conditions that influence landing outcomes.

Business Problem

Can we predict whether a SpaceX booster will land successfully **before launch**?

Accurate predictions would help:

- Estimate **true mission cost and profitability**
- Support **risk assessment and mission planning**
- Optimize decisions on **launch configuration and resource allocation**

Analytical Objective

We aim to build a classification model that:

- Learns from past SpaceX missions
- Identifies key factors driving landing success
- Predicts landing outcomes for future launches

Section 1

Methodology

Data Collection

Data Sources Used

- SpaceX REST API
- Wikipedia Falcon 9 launch tables

Purpose

- Collect historical Falcon 9 launch data
- Identify landing success outcomes
- Prepare dataset for prediction modeling

Tools Used

- Python
- requests library
- pandas
- BeautifulSoup

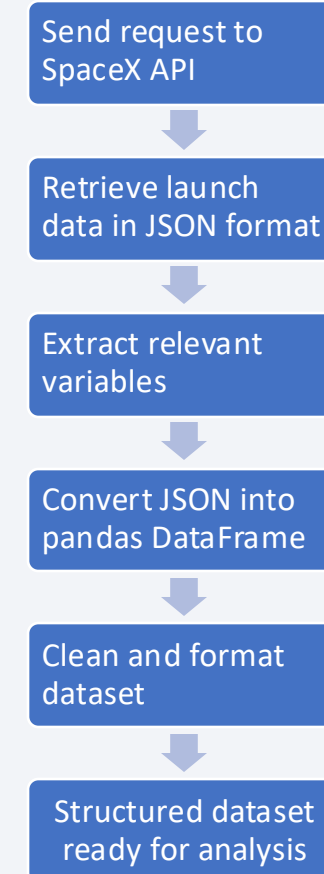
Data Collection – SpaceX API

Process

- Send request to SpaceX API
- Retrieve JSON launch data
- Extract mission variables
- Convert JSON to DataFrame
- Clean missing values

Variables Collected

- Flight number
- Launch date
- Booster version
- Payload mass
- Launch site
- Orbit type



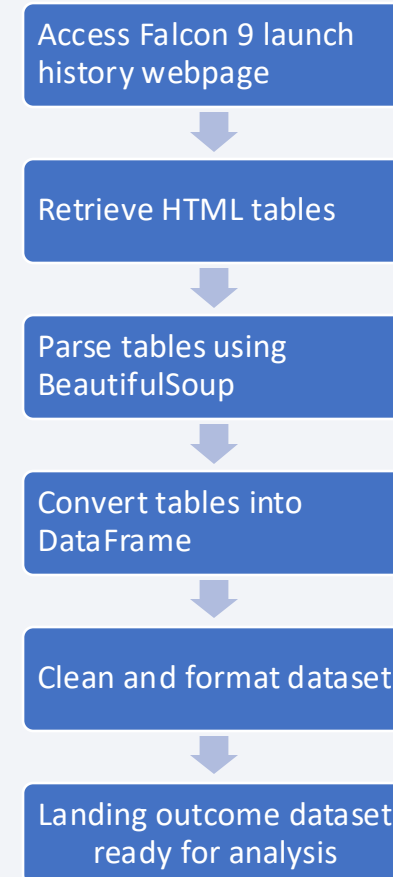
Data Collection - Scraping

Process

- Access Falcon 9 launch history page
- Extract HTML tables
- Parse tables using BeautifulSoup
- Convert tables → DataFrame
- Clean formatting issues

Variables Collected

- Booster landing outcome
- Landing method
- Mission results
- Reused booster information



Data Wrangling

Filtering Data

- Select Falcon 9 launches only
- Remove incomplete records
- Handling Missing Values
- Replace missing payload mass values

Feature Selection

Keep important variables:

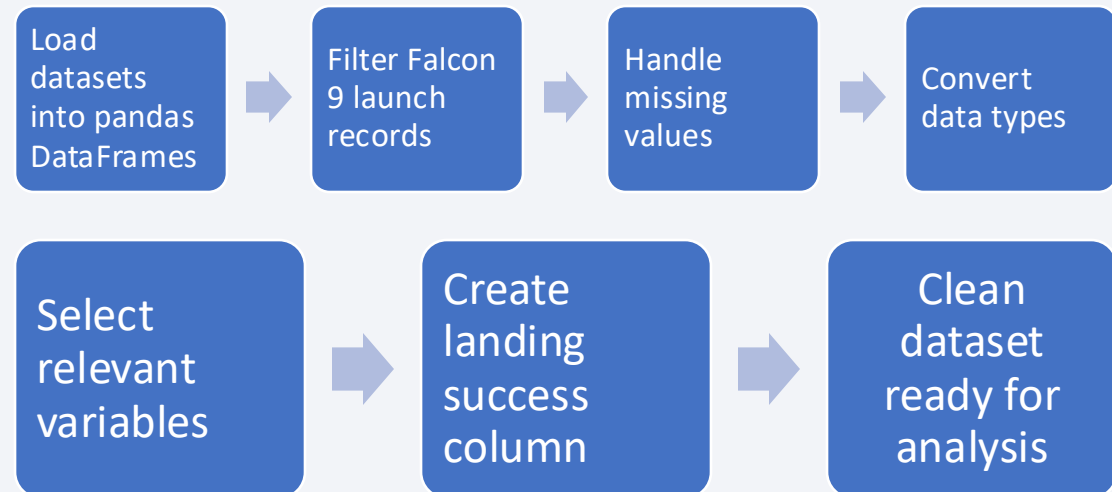
- Flight number
- Launch site
- Payload mass
- Orbit
- Booster version
- Landing outcome

Data Formatting

- Convert dates to datetime format
- Convert payload mass to numeric format

Landing Outcome Processing

- Extract landing result column
- Convert landing results into categories
- Create boolean landing success variable



EDA with Data Visualization

Methods

Compare data and create visualizations to show relationships between different variables, focusing on variables' relationships to class (success).

Variable Comparison

- Scatter Plots for display
- X and Y are the variables being compared
- Color of points is Class – representing outcome

Yearly Success Rate

- Line Graph to show trend based on time
- X is year and Y is success

EDA with SQL

Methods

- Loads a SpaceX dataset (CSV) into a database (Db2)
- Uses SQL queries to explore launch records
- Investigates key factors like payloads, launch sites, and booster landings

Key Insights

- Reusable first-stage boosters significantly reduce launch costs
- Predicting successful landings helps estimate mission expenses and competitiveness

Build an Interactive Map with Folium

Creating a map with folium helped to visualize what geographic features might impact the success or failure of a SpaceX launch.

- Adding colors to each site helped to visualize which sites were more successful
- Adding distance to highways/railroads, etc. Helped to determine what geographic and population features might impact launch success

Build a Dashboard with Plotly Dash

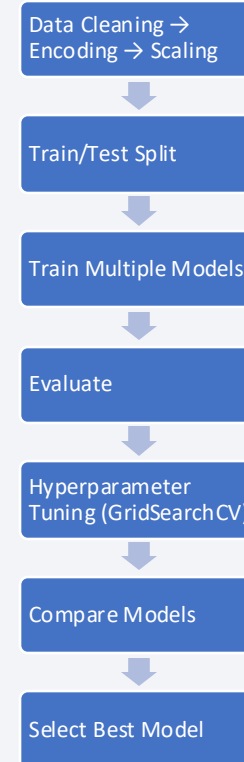
Creating a dashboard helped to visualize in realtime the data and how it is affected by different variables.

- Changing the Launch Site option helps to see specifically the success rate for each site clearly on the pie chart
- Changing the Payload Range variable helps to see the successes and failures of boosters within different ranges of payload

Predictive Analysis (Classification)

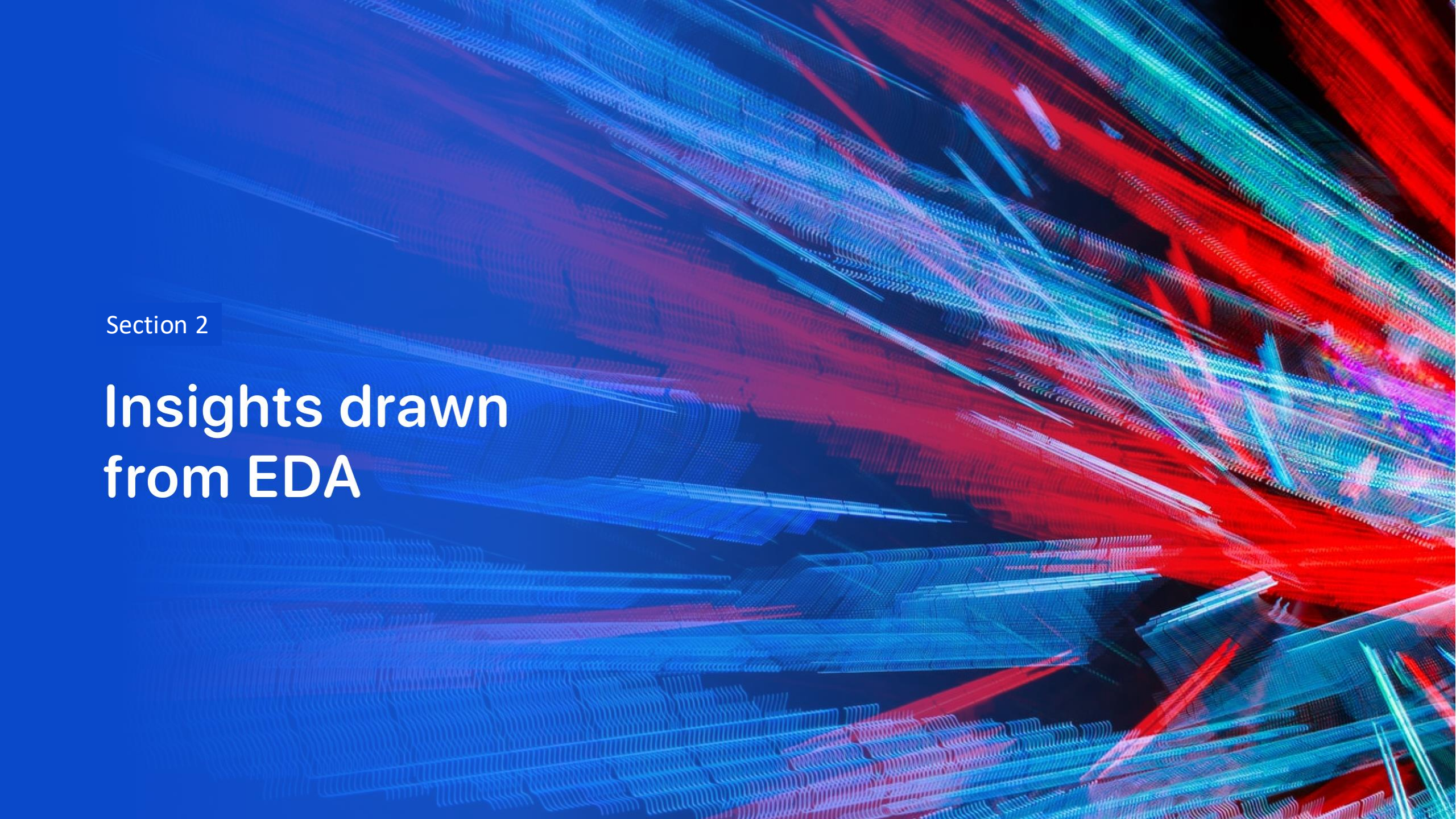
Building a Predictive Model:

- Inspected dataset structure
- Handled missing values
- Encoded categorical features
- Scaled numerical features
- Split data into training and testing sets



Results

- There are many strong correlations between the variables in the dataset. Many factors can affect the ability for the booster to land successfully.
- Building an interactive map and dashboard helped visualize the data at realtime and helps show correlations between specific variables.
- The Decision Tree model works best for determining whether or not a booster will land successfully

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of blue and red, creating a sense of motion or data flow. A faint, light blue grid pattern is also visible, particularly in the lower-left quadrant. The overall effect is high-tech and digital.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

What the chart shows

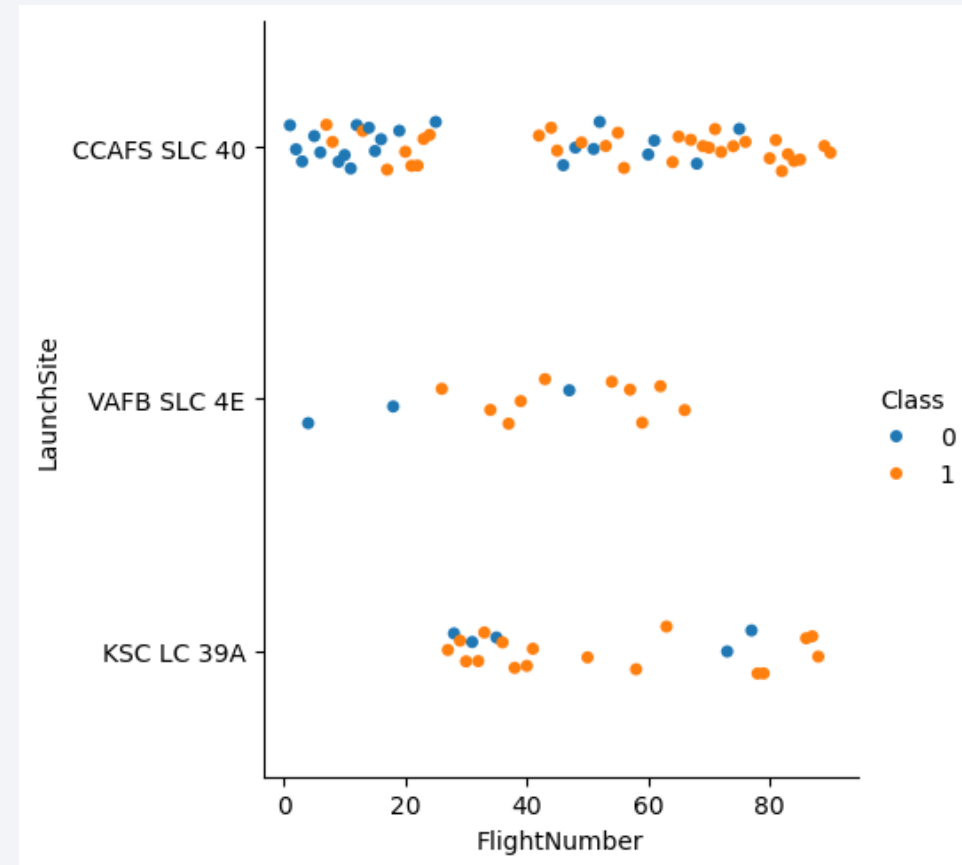
- Each point represents a SpaceX flight, plotted by its Flight Number (x-axis) and Launch Site (y-axis).
- Point color indicates Class:

Key patterns

- KSC LC-39A shows a large cluster of successful flights, especially at higher flight numbers.
- CCAFS SLC-40 has a mix of successes and failures, mostly in earlier flight numbers.
- VAFB SLC-4E has fewer total launches, but the points lean toward successful outcomes.

Overall takeaway

- As flight numbers increase, success rates improve, and KSC LC-39A appears to be the most consistently successful launch site.



Payload vs. Launch Site

What the chart shows

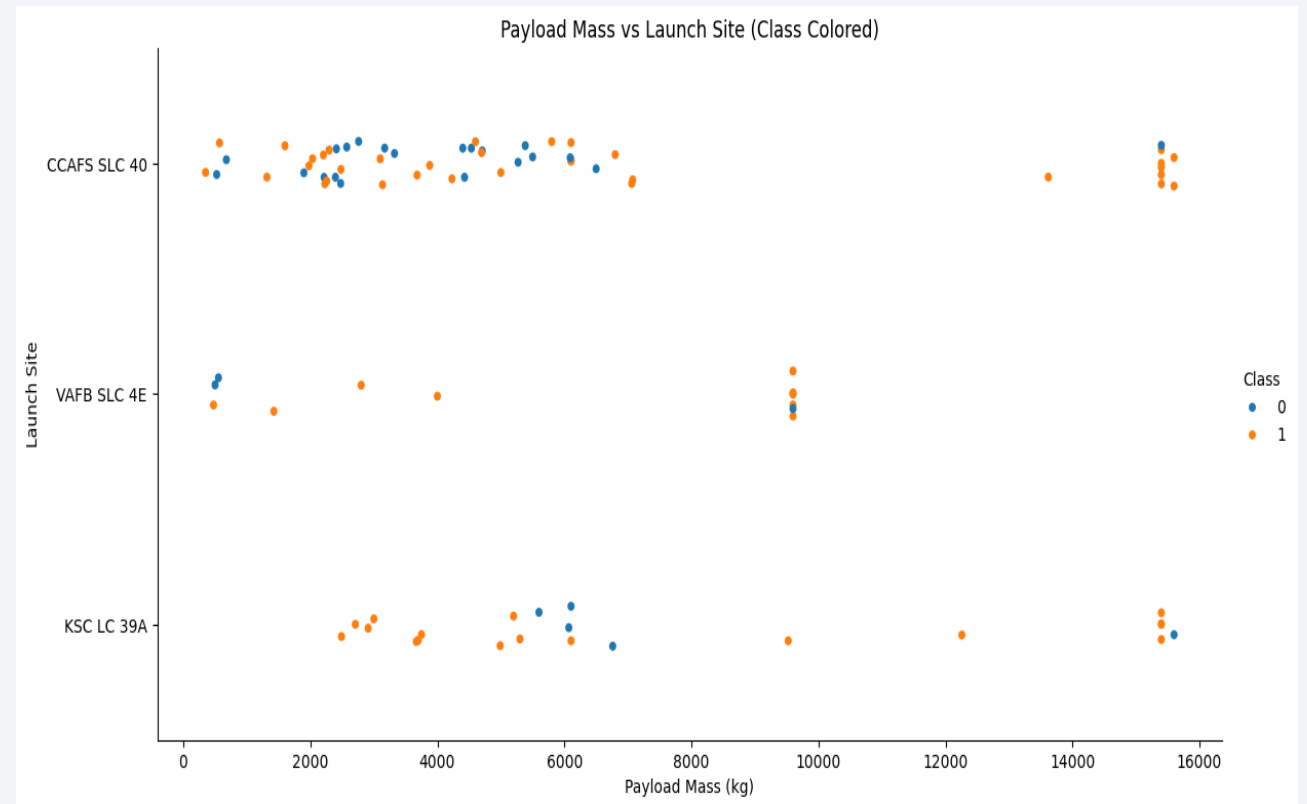
- Each point represents a SpaceX launch, plotted by Payload Mass (kg) on the x-axis and Launch Site on the y-axis.
- Point Color indicates Class:

Key patterns

- KSC LC-39A handles many of the heaviest payloads, with a strong concentration of successful launches at higher masses.
- CCAFS SLC-40 shows a wide spread of payload masses, but includes more lower-mass failures compared to the other sites.
- VAFB SLC-4E has fewer launches overall, mostly in the mid-range payload masses.

Overall takeaway

- Heavier payloads tend to launch from KSC LC-39A, and success rates improve at that site.
- Failures are more common at lower payload masses, especially from CCAFS SLC-40.



Success Rate vs. Orbit Type

What the chart shows

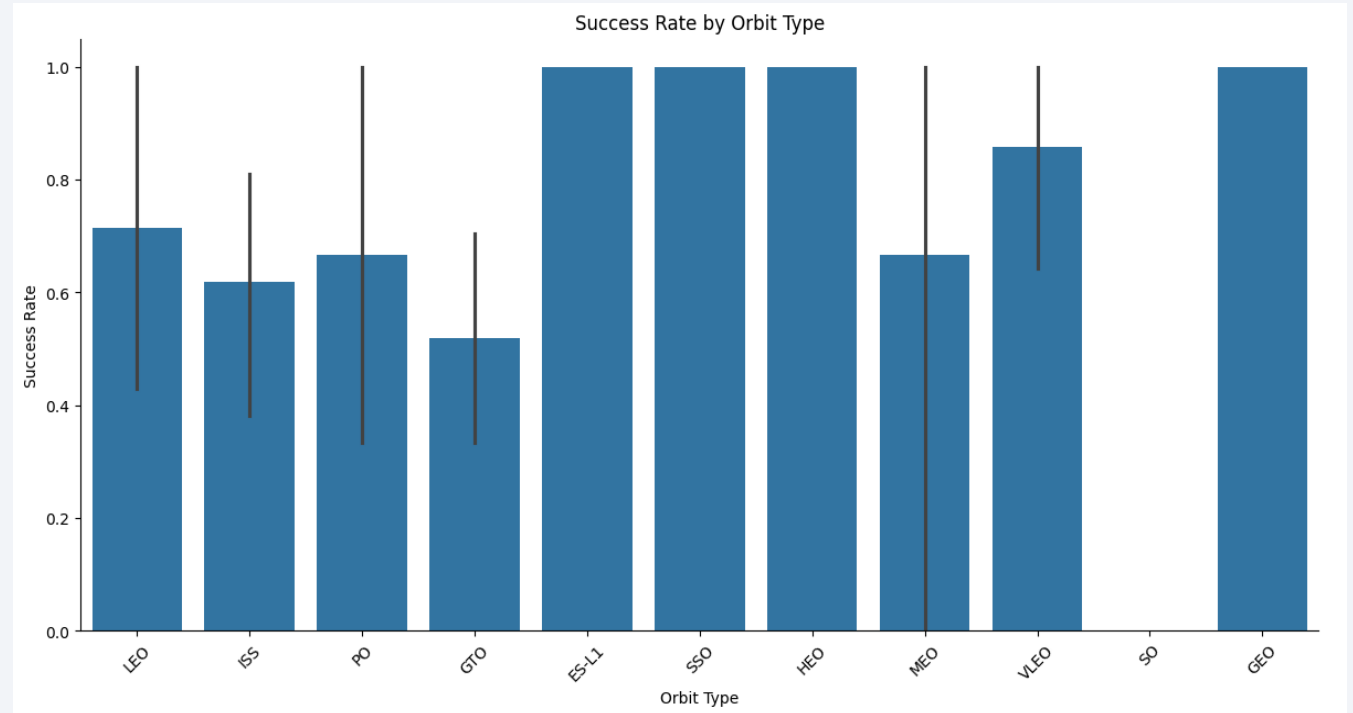
- Each point is a SpaceX flight, plotted by Flight Number (x-axis) and Orbit Type (y-axis).
- Color shows success class:

Key patterns

- LEO, ISS, and GTO have the largest number of missions, and later flights in these orbits are successful.
- Specialized orbits (ES-L1, HEO, MEO, GEO) have fewer flights, but the majority of those are successful.

Overall takeaway

- Across nearly all orbit types, success rates rise with experience, and later missions demonstrate consistent, high reliability.



Flight Number vs. Orbit Type

What the chart shows

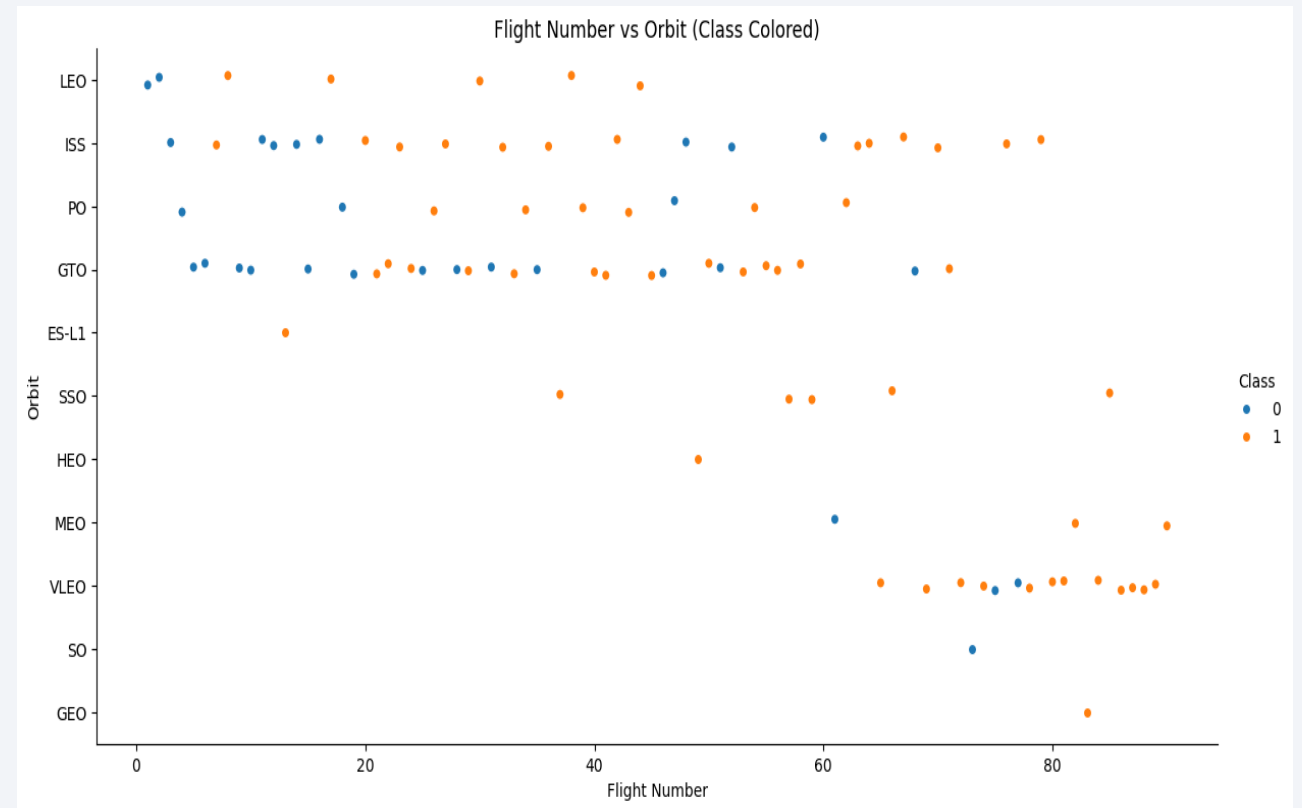
- Each point represents a SpaceX flight, plotted by Flight Number (x-axis) and Orbit Type (y-axis).
- Color indicates Class:

Key patterns

- Higher flight numbers show a strong shift toward successful missions across most orbit types.
- LEO, ISS, and GTO appear most frequently, with later missions in these orbits trending strongly successful.

Overall takeaway

- As SpaceX's flight experience increases, success rates rise across nearly all orbit types, with later missions showing consistent reliability.



Payload vs. Orbit Type

What the chart shows

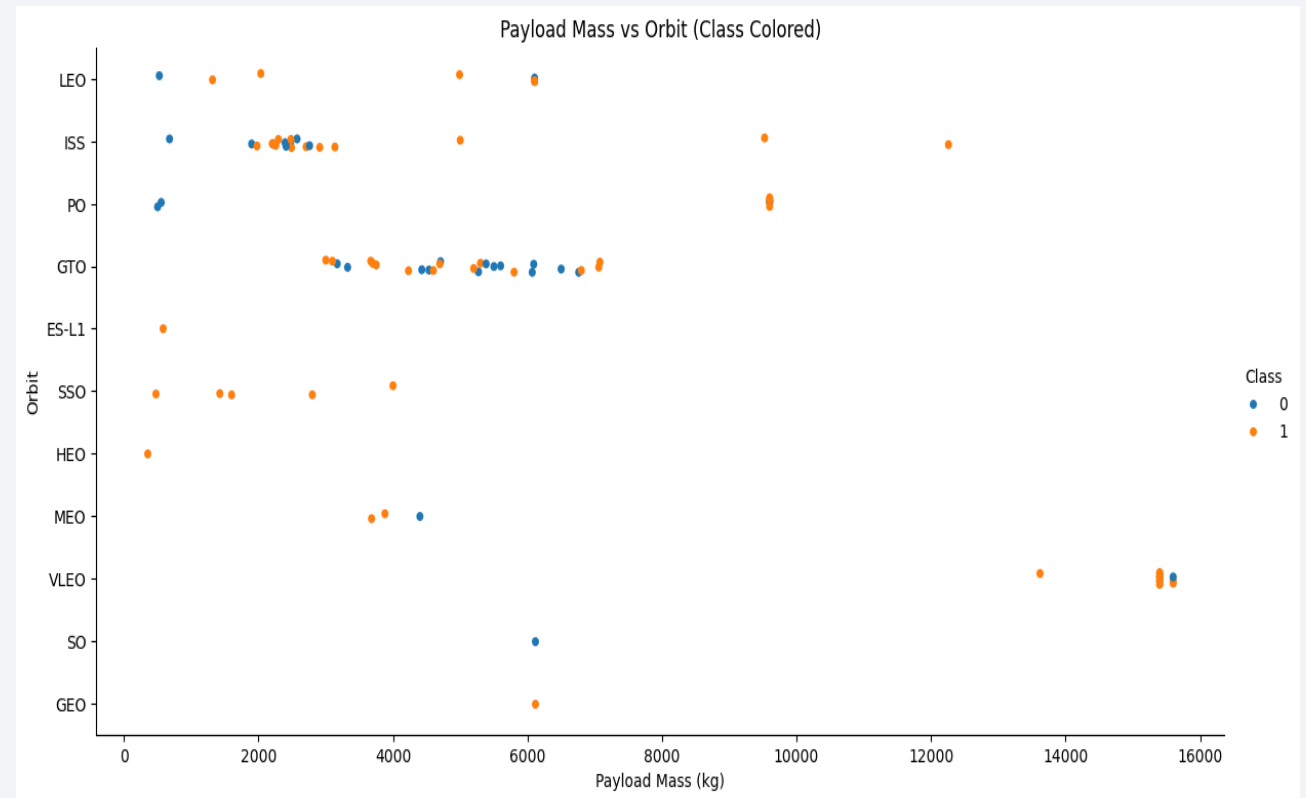
- Each point represents a SpaceX payload, plotted by Payload Mass (kg) on the x-axis and Orbit Type on the y-axis.
- Color indicates Class:

Key patterns

- Higher-mass payloads tend to correspond with successful missions, especially in GTO and LEO.
- Specialized orbits (ES-L1, HEO, MEO, GEO) show fewer total launches, but most points in these categories are successful.
- Lower-mass payloads show a mix of outcomes, with failures more common in the early or lower-mass range.

Overall takeaway

- Across orbit types, heavier payload missions are generally successful, and most orbits show strong reliability, especially in later or higher-mass flights.



Launch Success Yearly Trend

What the chart shows

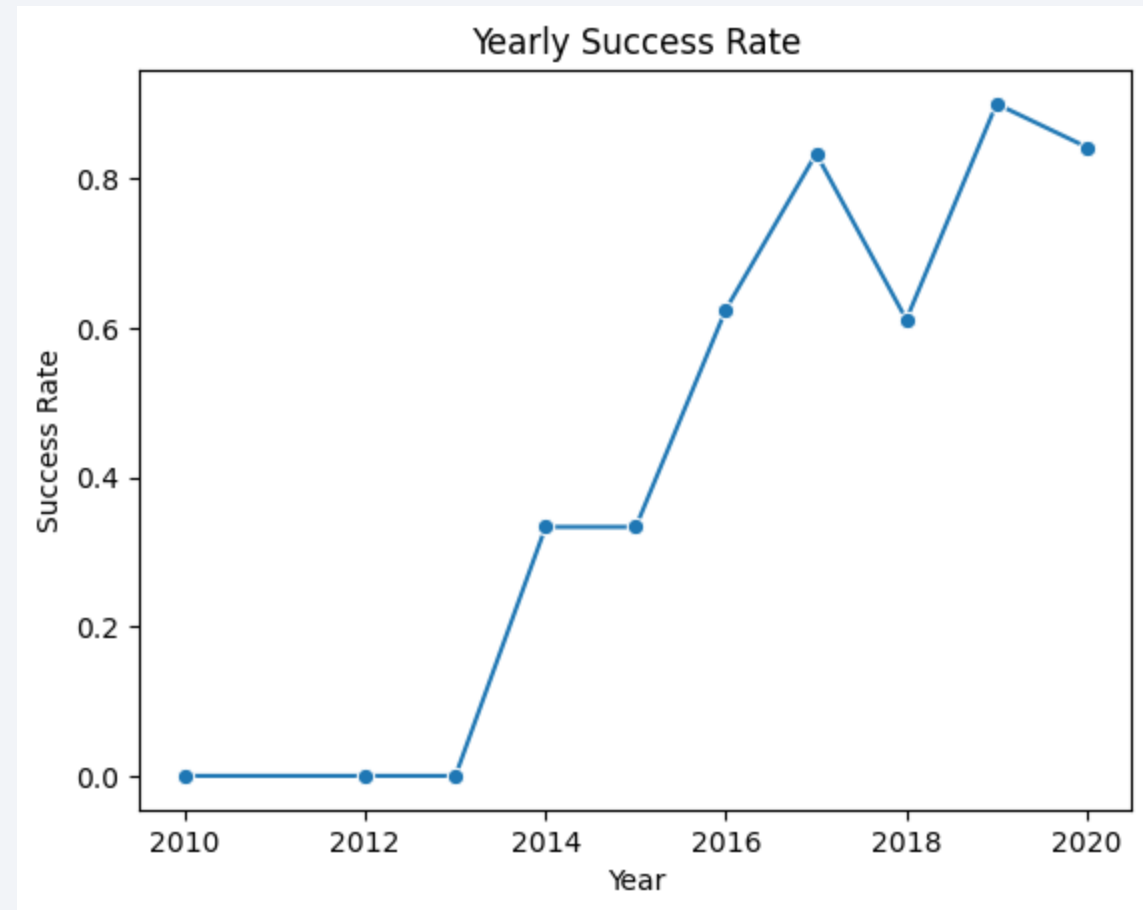
- A year-by-year success rate for SpaceX missions from 2010–2020.
- Values range from 0.0 to 1.0, showing the proportion of successful launches each year.

Key patterns

- 2010–2013: Success rate stays near zero, reflecting early development and testing phases.
- 2014–2017: A sharp rise, marking rapid improvements in reliability.
- 2018: A small dip interrupts the upward trend.
- 2019: The peak success rate, indicating the most reliable year in the dataset.
- 2020: Slight decline, but still high overall performance.

Overall takeaway

- Over the decade, SpaceX shows a clear upward trajectory in reliability, with early struggles turning into consistently strong performance in later years.



All Launch Site Names

All Launch Sites:

- CCAFS SLC 40
- VAFB SLC 4E
- KSC LC 39A

Launch Sites Beginning With 'CCA'

FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	Landing Pad	Block	ReusedCount	Serial	Longitude	Latitude	Class	Year
0	1	2010	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0003	-80.577366	28.561857	2010
1	2	2012	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0005	-80.577366	28.561857	2012
2	3	2013	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0007	-80.577366	28.561857	2013
4	5	2013	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1004	-80.577366	28.561857	2013
5	6	2014	Falcon 9	3325.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1005	-80.577366	28.561857	2014

Total Payload Mass

Total Payload Mass for all boosters:

619967 kg

Average payload mass for version F9 v1.1:

2928.4 kg

First Successful Ground Landing Date

First successful ground landing date:

December 22, 2015 (12-22-2015)

Successful Drone Ship Landing with Payload between 4000 and 6000

Successful Boosters that landed on drone ships with payload mass between 4000 and 6000:

- B1022
- B1026
- B1021
- B1040
- B1031
- B1032
- B1046
- B1059

Total Number of Successful and Failure Mission Outcomes

- Total Successful Missions:
- 60
- Total Failed Missions:
- 30

Boosters Carried Maximum Payload

Boosters that carried max payload:

- F9 B5 B1022
- F9 B5 B1026
- F9 B5 B1021
- F9 B5 B1040
- F9 B5 B1031
- F9 B5 B1032
- F9 B5 B1046
- F9 B5 B1059

2015 Launch Records

- Failed landing_outcomes in drone ship for year 2015

BoosterVersion	LaunchSite	LandingPad	Outcome	Date
11	Falcon 9	CCAFS SLC 40	5e9e3032383ecb761634e7cb	2015
13	Falcon 9	CCAFS SLC 40	5e9e3032383ecb761634e7cb	2015
14	Falcon 9	CCAFS SLC 40	NaN	2015
15	Falcon 9	CCAFS SLC 40	5e9e3032383ecb6bb234e7ca	2015

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Count of landing outcomes between the date 2010-06-04 and 2017-03-20:

- Success: 24
- Failures: 18

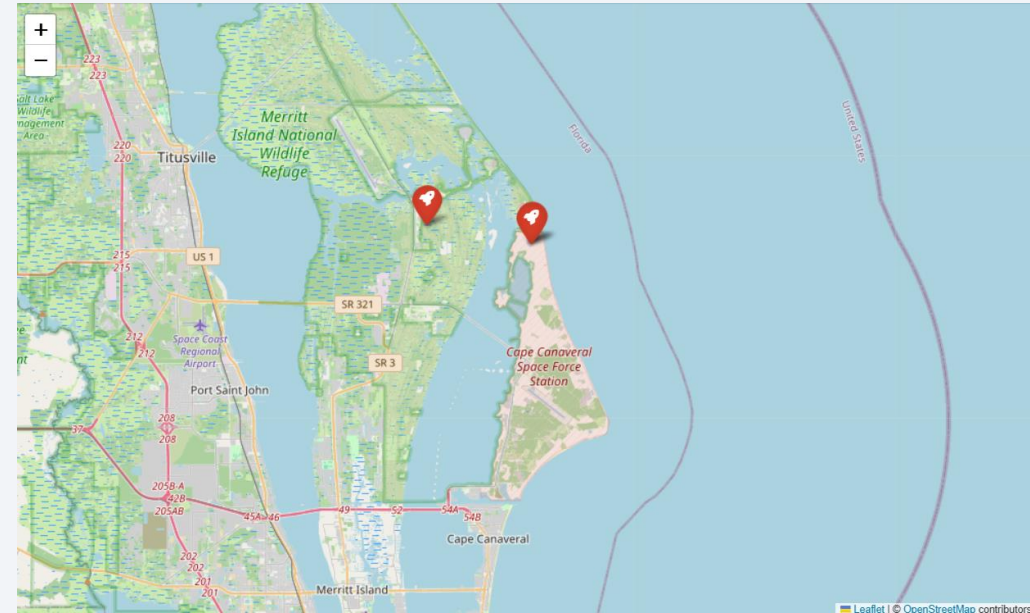
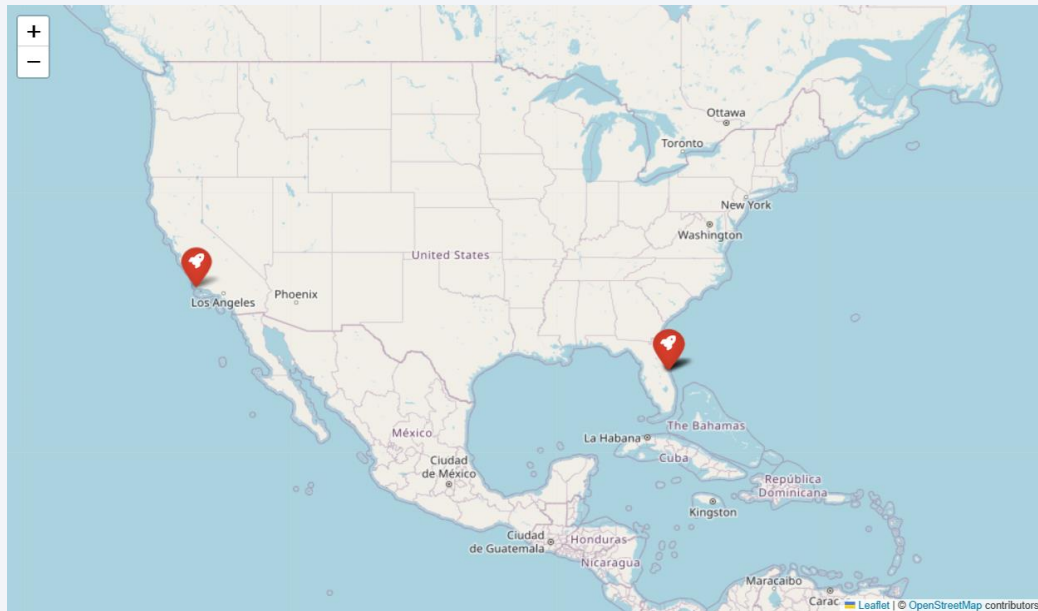
A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a solid blue background on the left and a satellite photograph of Earth on the right. The Earth's surface is dark, with numerous bright yellow and orange lights representing cities and urban areas. The horizon of the Earth is visible as a thin, curved line separating the dark surface from the deep blue of space.

Section 3

Launch Sites Proximities Analysis

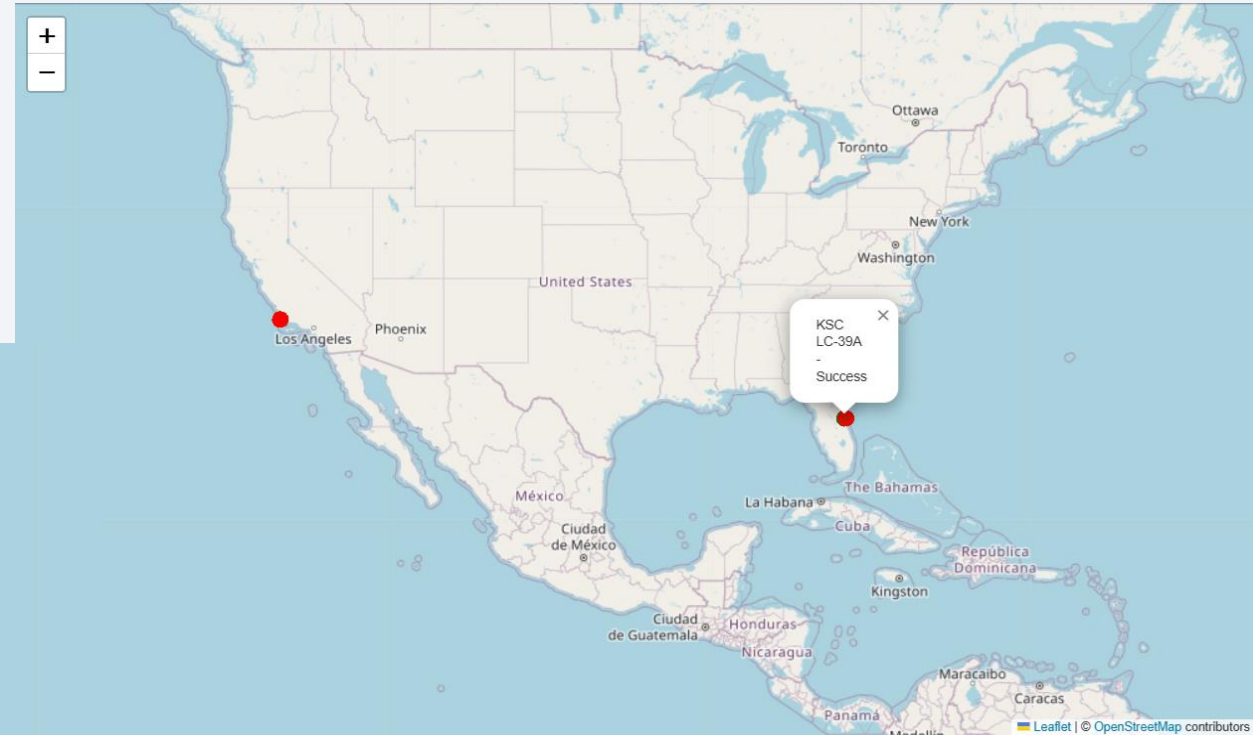
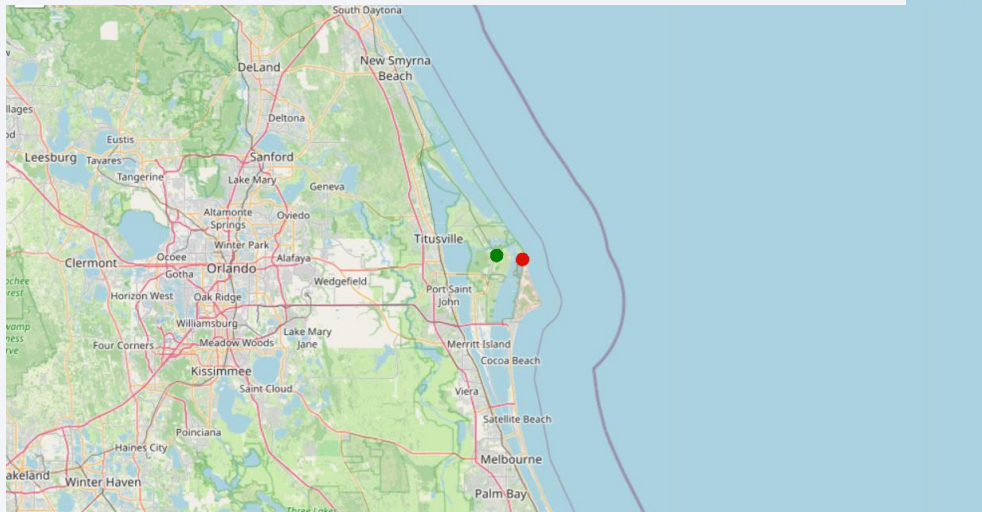
Folium Map – All Launch Site Locations

This map marks all Launch Site locations in the SpaceX dataframe:



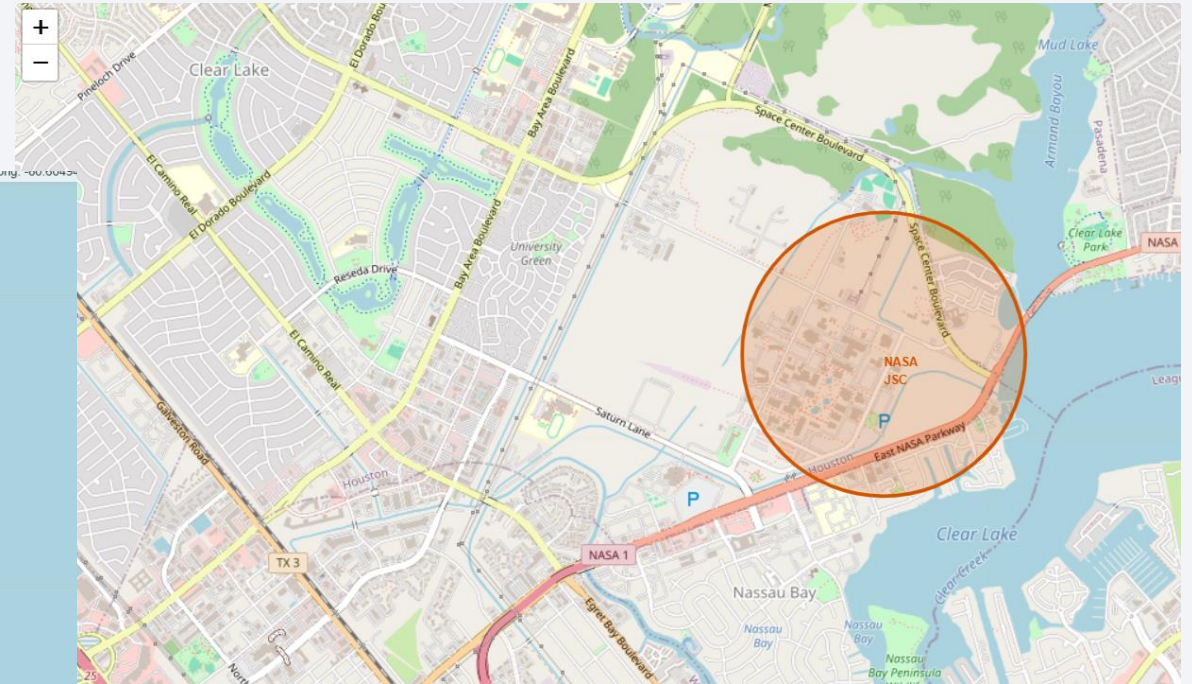
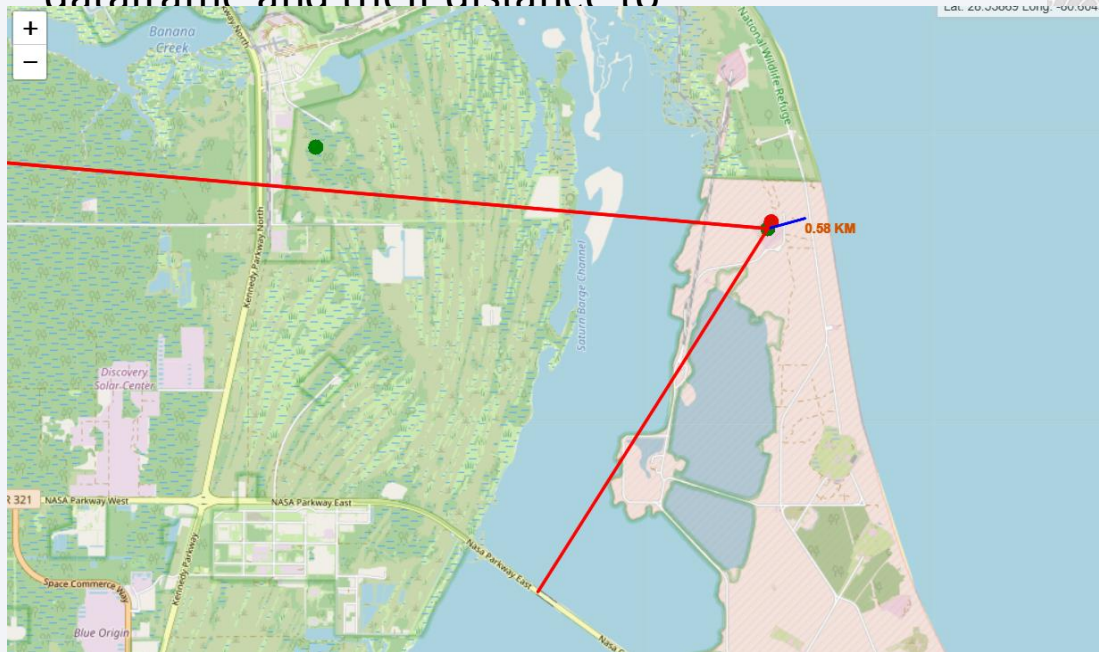
Folium Map – Success/Fail Map

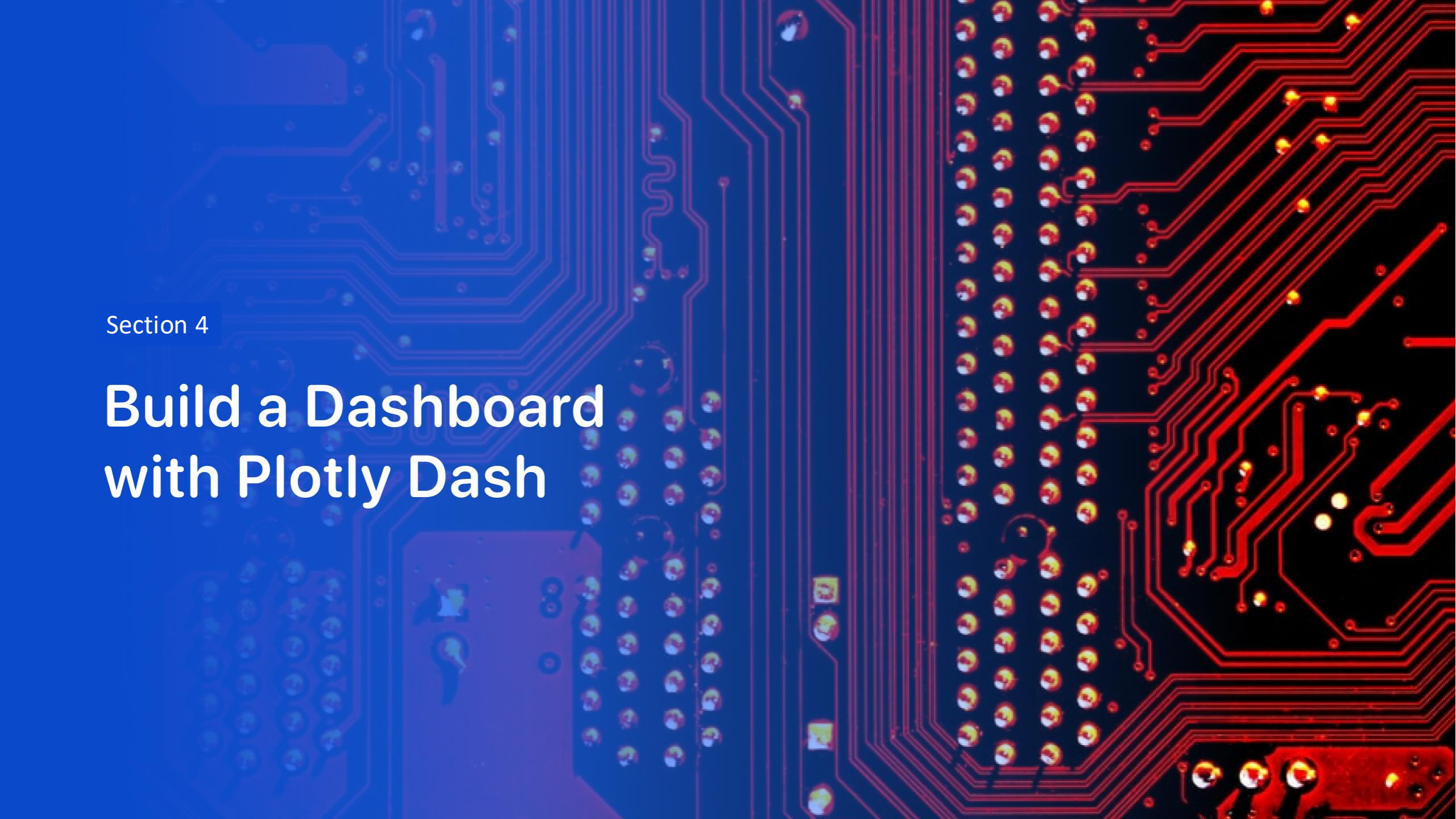
This map marks all Launch Site locations in the SpaceX dataframe based on successful or unsuccessful launch (red/green):



Folium Map – Distance Markers

This map marks all Launch Site locations in the SpaceX dataframe and their distance to



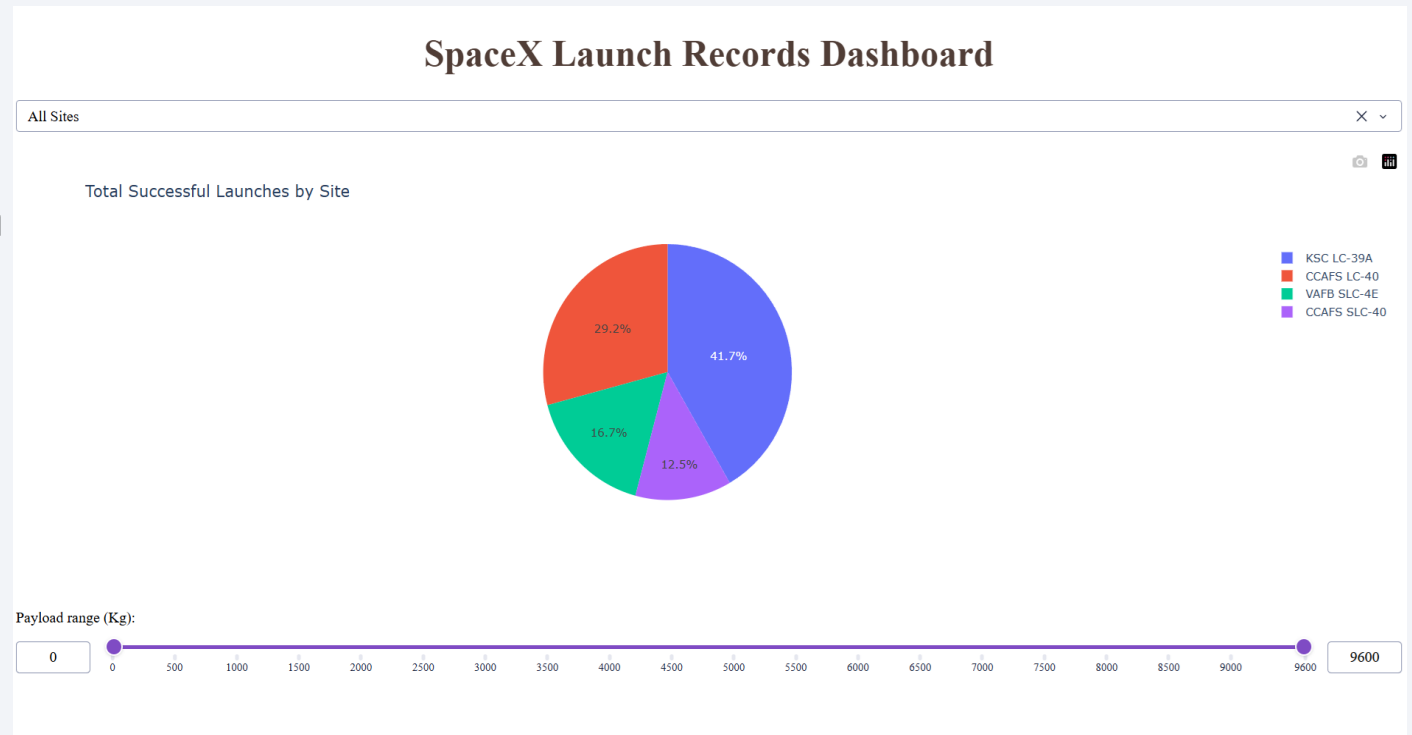


Section 4

Build a Dashboard with Plotly Dash

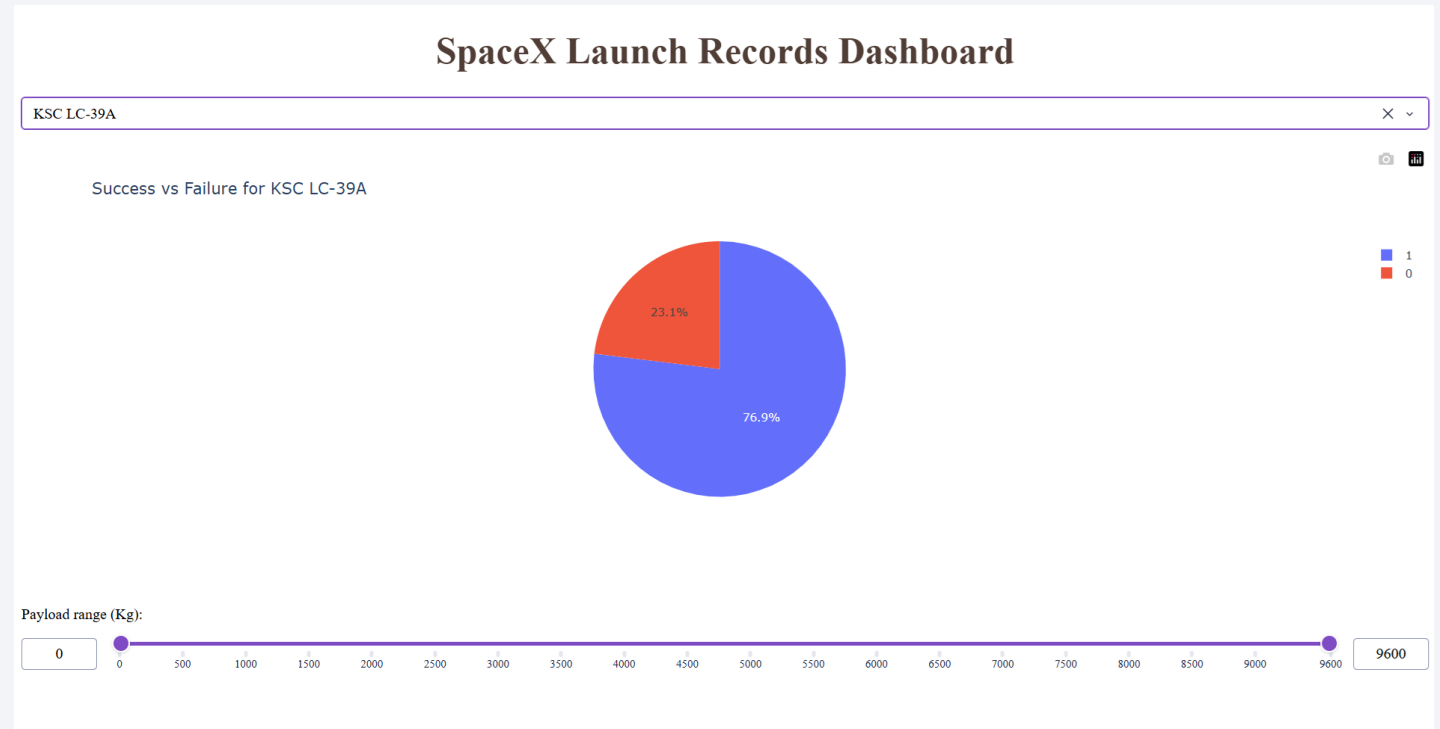
Dashboard 1: Success Count for All Launch Sites

- The pie chart shows total successful launches by site (site being the color shown in the legend)
- The slider at the bottom shows the payload range.



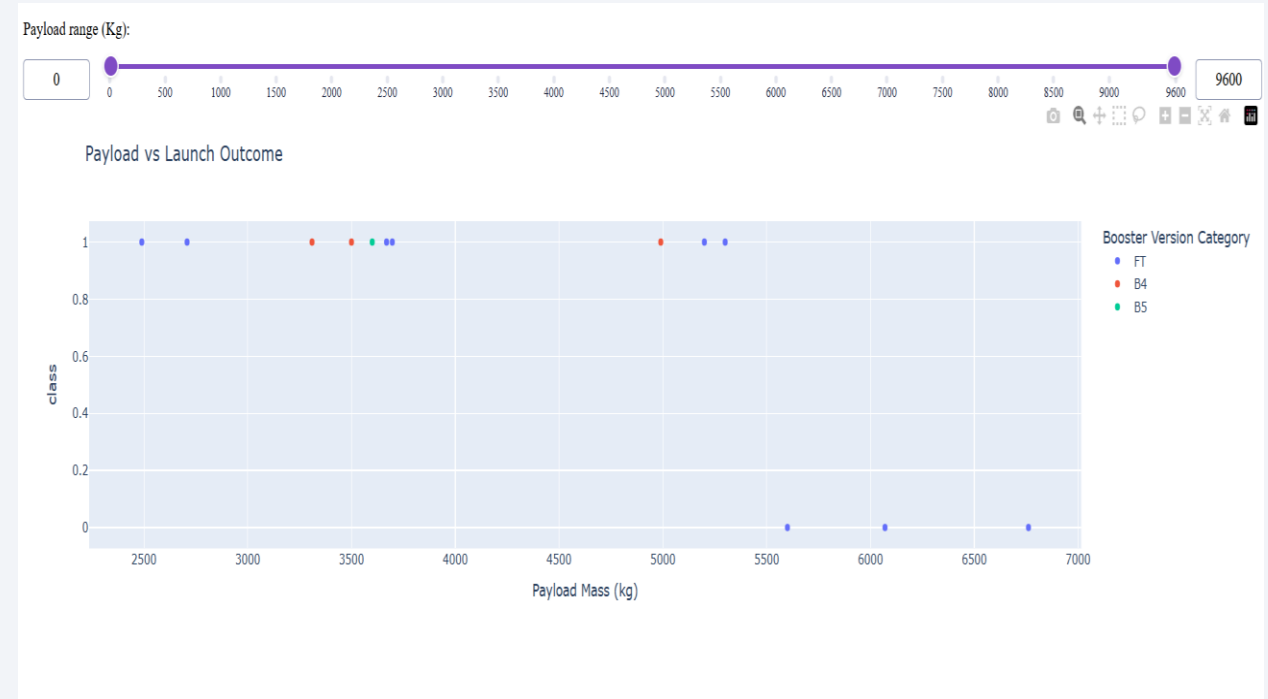
Dashboard 2: Success Rate of Most Successful Site

- The pie chart shows the ratio of successful to unsuccessful launches for the most successful site.
- The slider at the bottom shows the payload range.



Dashboard 3: Payload Vs Launch Outcome

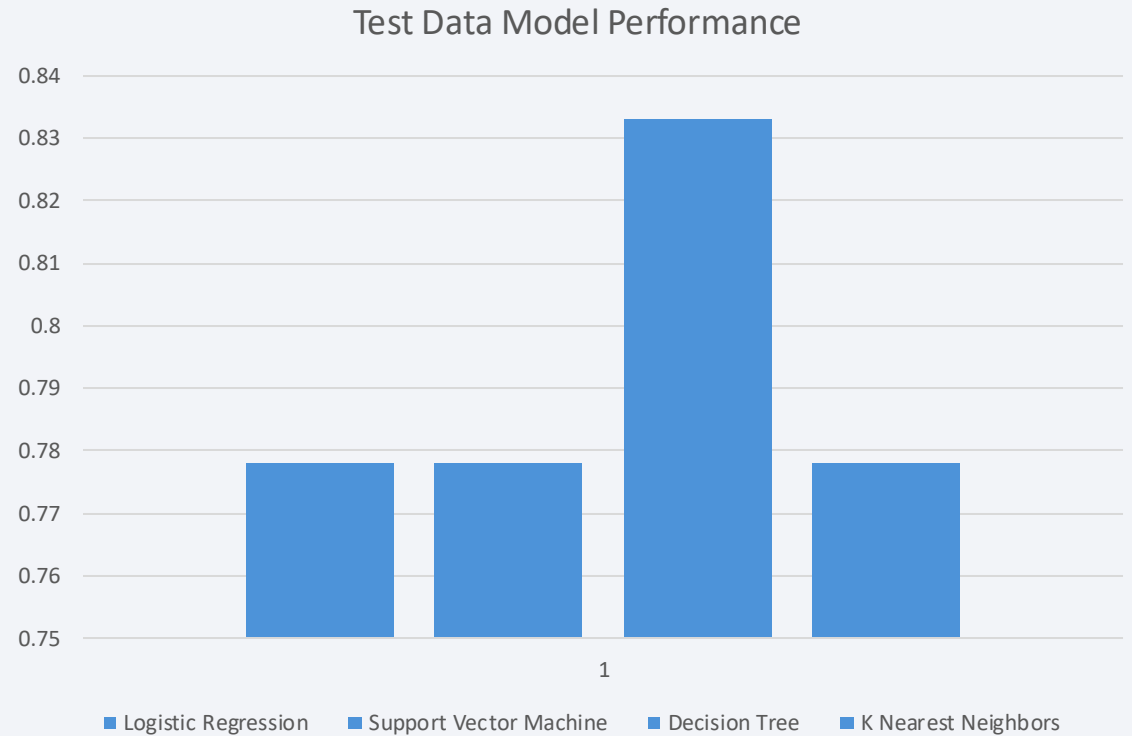
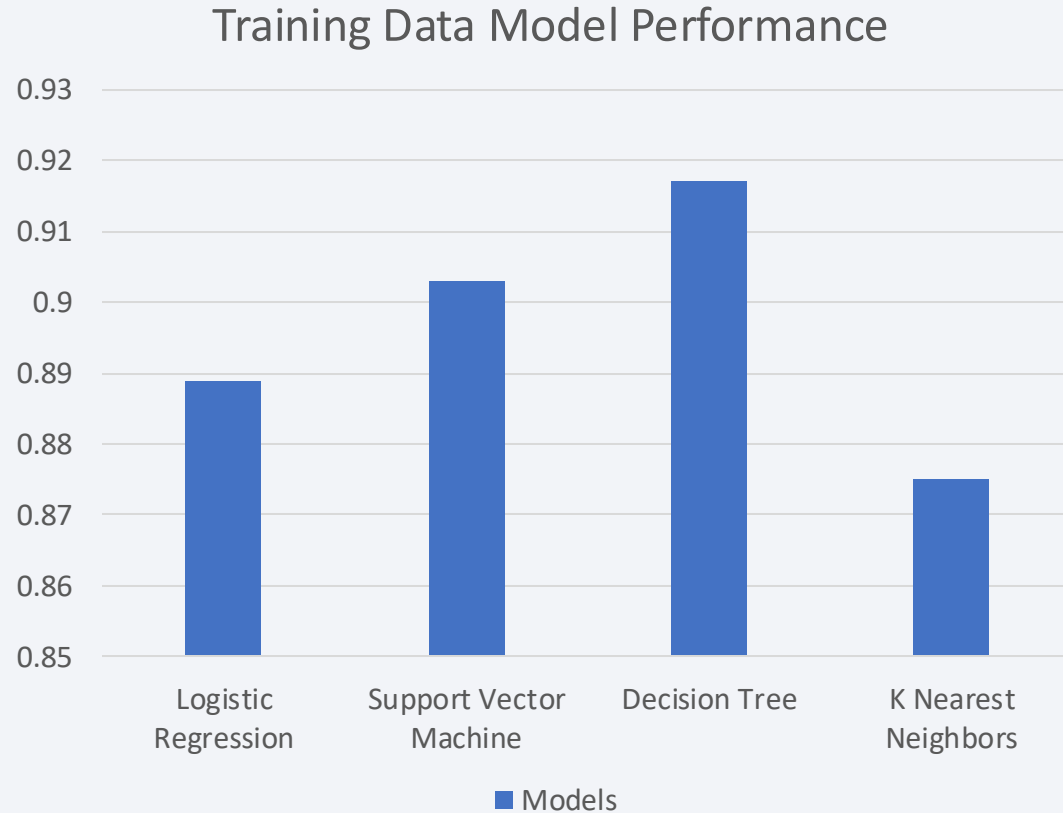
- The slider at the top shows the payload range.
- The points on the chart represent launches with color showing booster type.



Section 5

Predictive Analysis (Classification)

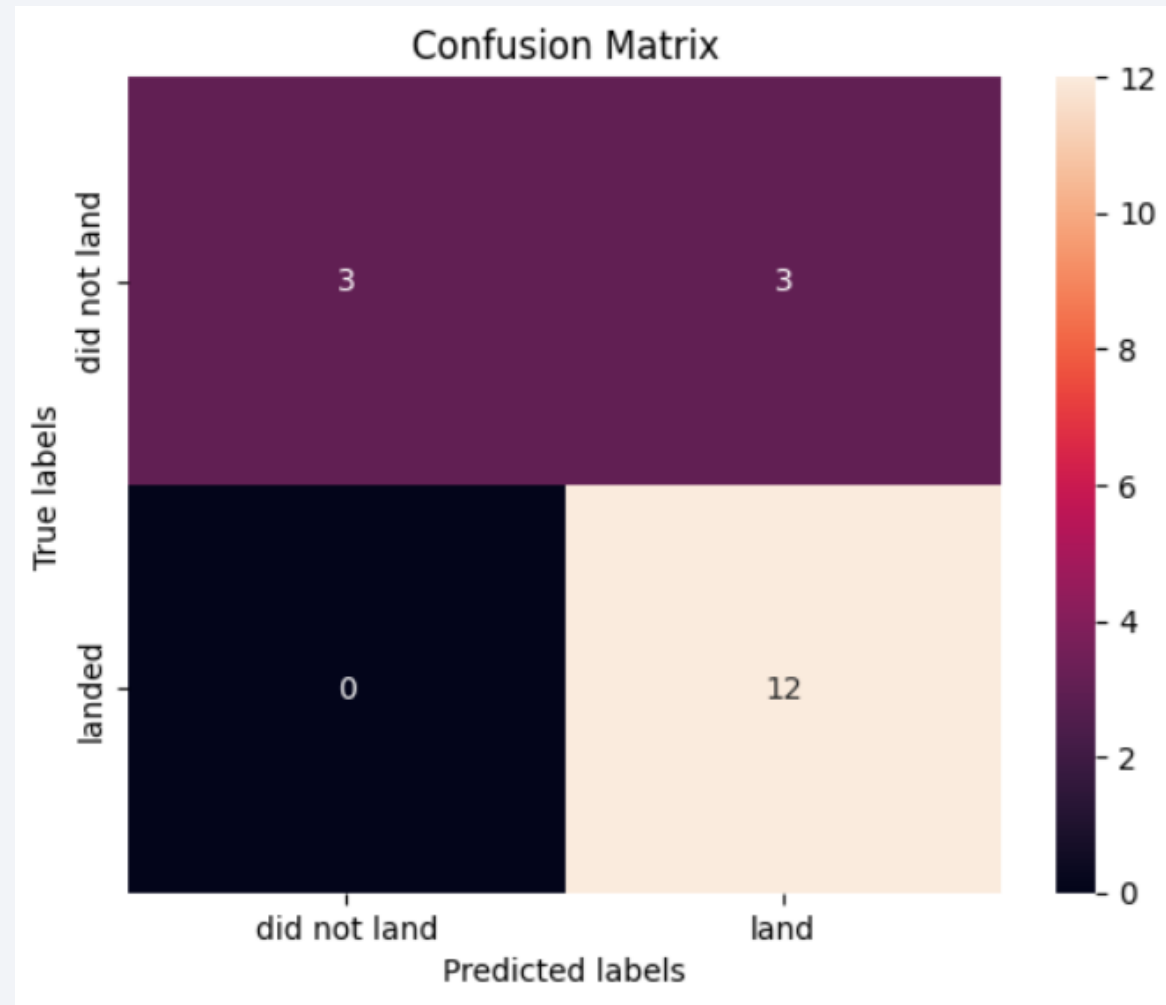
Classification Accuracy



Confusion Matrix

Best Performing Model:

Decision Tree



Conclusions

- The Decision Tree model works best for determining whether or not a booster will land successfully. It has a very high true positive rate.
- Using this model, we can with fairly good accuracy, predict the outcome of a SpaceX Launch.
- Using such methods to predict landing outcomes can significantly decrease SpaceX's launch costs and success rates. By analyzing the factors that make a launch successful or unsuccessful, SpaceX can tailor their launches to be as profitable as possible.

Appendix

- GitHub Link to all completed notebooks:

https://github.com/Unitymaster50/Data_Science_Capstone_Repo.git

Thank you!

